



## Conclusion

## Summary of Contributions

This paper provided comprehensive empirical validation of the Cognitive Integrity Framework (CIF) introduced in Part 1 of this series. Our primary contributions:

**Implementation:** We implemented the complete CIF defense suite—cognitive firewalls, belief sandboxes, trust calculus with bounded delegation, tripwire detection, behavioral invariants, and Byzantine-tolerant consensus—demonstrating that the formal mechanisms translate into deployable code.

**Attack Corpus:** We assembled 950 cognitive attacks across four categories (prompt injection, trust exploitation, belief manipulation, coordination attacks), enabling reproducible security evaluation of multiagent systems.

**Cross-Architecture Validation:** We evaluated CIF across six production multiagent architectures (Claude Code, AutoGPT, CrewAI, LangGraph, MetaGPT, Camel), demonstrating that formal guarantees hold across diverse architectural patterns.

# Key Findings

1. **Layered defense is essential:** No single mechanism achieves acceptable protection; composition yields multiplicative improvement consistent with theoretical predictions.
2. **Trust calculus prevents amplification:** The  $\delta^d$  decay bound successfully prevented trust laundering across all tested architectures—a structural guarantee independent of attacker sophistication.
3. **Architecture matters:** Peer-to-peer architectures show greatest improvement from CIF, consistent with their vulnerability to lateral movement attacks.
4. **Performance overhead is acceptable:** 20-25% latency overhead for full CIF deployment is appropriate for security-critical contexts.

# Implications for Practitioners

The empirical results validate that CIF provides practical protection:

- ▶ **Deploy layered defenses:** Configure all CIF components for security-critical deployments
- ▶ **Calibrate to architecture:** Apply architecture-specific recommendations from `tab:architecture-insights`
- ▶ **Monitor continuously:** Detection rates degrade over time; ongoing vigilance is required

For detailed deployment guidance, including human-actionable checklists and agent-readable guidelines, see Part 3 of this series.

# Paper Series

This is Part 2 of the *Cognitive Security for Multiagent Operators* series:

- ▶ **Part 1: Formal Foundations** - Trust calculus, defense composition algebra, information-theoretic bounds
- ▶ **Part 2 (This Paper): Computational Validation** - Implementation, attack corpus, empirical results
- ▶ **Part 3: Practical Guidance** - Deployment checklists, operator posture, risk assessment

Together, these papers provide a complete framework for understanding, implementing, and operating cognitive security in multiagent AI systems.